@articleross$_i$nterplay$_2$024, title = Theinterplaybetweendietandthegutmicrobiome : implicationsfo
22, issn = 1740 − 1534, doi = 10.1038/s41579 − 024 − 01068 − 4, shorttitle =
Theinterplaybetweendietandthegutmicrobiome, abstract = Diethasapivotalroleinshapingthecomposit
671 − −686, number = 11, journaltitle = NatureReviews.Microbiology, shortjournal =
NatRevMicrobiol, author = Ross, FionaC.andPatangia, DhratiandGrimaud, GhjuvanandLavelle, A
2024 − 11, pmid = 39009882, keywords = Animals, Diet, Diet, Ketogenic, Diet, Mediterranean, Gast

@articlearmet$_r$ethinking$_2$022, title = Rethinkinghealthyeatinginlightofthegutmicrobiome, volume
30, issn = 1934 − 6069, doi = 10.1016/j.chom.2022.04.016, abstract = Giventheworldwideepidemicof
764 − −785, number = 6, journaltitle = CellHost&Microbe, shortjournal =
CellHostMicrobe, author = Armet, AnissaM.andDeehan, EdwardC.andO′Sullivan, AidanF.andMo
2022 − 06 − 08, pmid = 35679823, keywords = Diet, Diet, Healthy, dietaryfiber, dietaryguidelines, G
FullText : CUsersaPodoZoterostorageDU8CETGCArmetetal. − 2022 − Rethinkinghealthyeatingi

@articlecresci$_g$ut$_2$015, title = GutMicrobiome : WhatWeDoandDon′tKnow, volume =
30, issn = 1941 − 2452, doi = 10.1177/0884533615609899, shorttitle = GutMicrobiome, abstract =
Withinthelastdecade, researchregardingthehumangutmicrobiomehasexploded.Whilethegastrointest
734 − −746, number = 6, journaltitle = NutritioninClinicalPractice : OfficialPublicationoftheAm
NutrClinPract, author = Cresci, GailA.andBawden, Emmy, date = 2015 − 12, pmid =
26449893, pmcid = PMC4838018, keywords = antibiotics, chronicdisease, diet, GastrointestinalMic
AcceptedVersion : CUsersaPodoZoterostorage2QDFXAVHCresciandBawden − 2015 − GutMicro

@articleheintz-buschart$_h$uman$_2$018, title = HumanGutMicrobiome : FunctionMatters, volume =
26, issn = 1878 − 4380, doi = 10.1016/j.tim.2017.11.002, shorttitle = HumanGutMicrobiome, abstrac
Thehumangutmicrobiomerepresentsacomplexecosystemcontributingessentialfunctionstoitshost.Re
563 − −574, number = 7, journaltitle = TrendsinMicrobiology, shortjournal =
TrendsMicrobiol, author = Heintz − Buschart, AnnaandWilmes, Paul, date =
2018 − 07, pmid = 29173869, keywords = functionalmicrobiome, GastrointestinalMicrobiome, Gast

@articlezahavi$_b$acterial$_2$023, title = BacterialSNPsinthehumangutmicrobiomeassociatewithhost
29, issn = 1546 − 170X, doi = 10.1038/s41591 − 023 − 02599 − 8, abstract =
Genome − wideassociationstudies(GWASs)haveprovidednumerousassociationsbetweenhumansingl
2785 − −2792, number = 11, journaltitle = NatureMedicine, shortjournal =
NatMed, author = Zahavi, LironandLavon, AmitandReicher, LeeandShoer, SaarandGodneva, Anas
2023 − 11, pmid = 37919437, pmcid = PMC10999242, keywords = Bacteria, BodyMassIndex, Gastr
PubMedCentralFullTextPDF : CUsersaPodoZoterostorageQXT23SK6Zahavietal. − 2023 − Bac

@onlinenoauthor$_3$132$_n$odate, title = 3.13.2Documentation, url = https : //docs.python.org/3/, url
2025 − 03 − 28, file = 3.13.2Documentation : CUsersaPodoZoterostorage4EENQHVT3.html : tex

@onlinenoauthor$_s$cikit−learn$_n$odate, title = scikit − learn : machinelearninginPython|scikit − le
https : //scikit − learn.org/stable/, urldate = 2025 − 03 − 28, file = scikit − learn machinelearning

citations.bib

Assignment 1

Name: Aris Podotas

University: National and Kapodistrian University of Athens

Program: Data Science and Informaion Technologies

Specialization: Bioinformatics - Biomedical Data

Lesson: Machine Learning In Computational Biology

Date: April 2025

Contents

1 Abstract

In recent years the role of the gut microbiome in health [?] has been elucidated.

2 Introduction

In this assignment we try to compare different *regressors* for *BMI* depending on the prevelance of and diversity of species of the gut microbiome.

3 Methods

3.1 Implementation

All results were created in the *Python* programming language [?].

3.2 Models

Three different *regressors* were used in the analysis.

Models:

1. Bayesiand Ridge
2. Support Vector Regression
3. Elastic Net

Models were imported from the *sklearn* [?] library in *Python* [?].

3.3 Evaluation metrics

Four evaluation metrics were chosen

1. Root mean squared error
2. Mean absolute error
3. R2 score
4. Median absolute error

The Root mean squared error, Mean absolute error, R2 score were chosen as evaluation metrics because they are listed as mandatory and the Median absolute error because we have seen the distributions of the pairs of features and we know:

1. The categorical data is not going to be in the final data, but in the final metadata.
2. Some sets of features contain what seem like outliers.
3. We have no median based metric till now to compensate

As a result of knowing about outliers a median metric must be included to compare with the non-median metrics.

3.4 Confidences

1000 iteration *bootstrapping* was done on models for evaluation along with repeated *k-fold cross validation* for 50 folds, 20 iterations.

The *sklearn* library [?] does not contain a *bootstrapping* function that we can use to call on our models. A protocol was written at our source files.

All files can be found at this repository.

4 Results

5 Disclaimer

Large language models (LLM's) were used during the assignment.

5.1 LLM

Open AI: Chat GPT 4.0 [**?**]

5.2 Purpose

1. To check ideas, like if the z-scaled data must be zero mean and unit variance (apparently it is) or if it's conditional (or to explain ideas).

2. To ask if concepts already exist in some dependency, as was the case for the way we cast the labels of the sex feature in the original data to binary representations (also re-used for the Project ID).

3. For acquisition of the relative documentation.

4. For helping with the LaTeXtable formating in this report for time efficiency.

6 Citations